

Temporal-Scale Modeling and Audio-Text Alignment for Audio Moment Retrieval with Zero-Training Saliency Reuse

Seungdeok Choi, Yong-Hwa Park

KAIST, Daejeon, Republic of Korea
 {haroldchoi6, yhpark}@kaist.ac.kr

Abstract—Audio moment retrieval (AMR) grounds free-form text queries as temporal spans in audio recordings. Target events range from short transients to tens-of-seconds scenes, so the tolerance of the tight-IoU metric varies substantially across queries. We address this with two design principles: modeling across temporal scales and aligning audio and text at multiple granularities. Our detector builds on QD-DETR over a single frozen M2D-CLAP encoder. For temporal scale, a coarse auxiliary branch supervises the shared encoder at halved resolution, denoising query training restores perturbed spans, a single-layer bidirectional Mamba encoder carries minute-long context, and a bounded cascade decoder re-attends to localized regions. For alignment, span re-ranking and boundary-contrastive objectives tie span and boundary regions to the query text, while a frame-level saliency objective aligns individual frames. At inference, zero-training saliency reuse converts this trained saliency head into a parallel proposal stream for proposal generation. Submitted to DCASE 2026 Task 6 as four cumulative variants, the system ranked joint sixth of 21 teams with 8.0 M trainable parameters per model. Paired per-query significance tests with false-discovery-rate control show that each stage improves at least one metric significantly, while several single-stage R1@0.7 differences remain within seed variability. Stratified analysis further shows that the largest gains occur for short, many-moment queries, where proposal coverage and boundary precision become tightly coupled at one-second temporal resolution.

Index Terms—Audio moment retrieval, temporal grounding, DETR, saliency, DCASE 2026

1. INTRODUCTION

Audio moment retrieval (AMR) aims to locate the start and end timestamps of audio segments that match a free-form text query within long, untrimmed recordings [1]. Although large-scale datasets such as AudioSet [2] and AudioCaps [3] have substantially advanced computational audio understanding, audio benchmarks have commonly formulated recordings as short, pre-segmented clips for tasks such as sound event detection [4], sound event localization [5], audio tagging [6], and clip-level audio–text retrieval [7]. This clip-centric formulation is well suited to recognizing what is contained in an isolated segment, but it does not address where a described sound occurs within the original recording. AMR directly targets this access problem by grounding the query as one or more temporal spans, enabling users to retrieve specific events from long-form audio such as broadcasts, surveillance recordings, and sound archives without manually reviewing the full recording.

The main challenge is not only determining whether the queried sound is present, but also identifying its temporal extent accurately. This requires reasoning at different temporal scales: short transients depend on fine-grained boundary cues, whereas extended acoustic scenes require broader context to distinguish the relevant interval from surrounding content [8], [9]. This disparity becomes especially pronounced under strict overlap-based evaluation. For an equal-length prediction shifted by δ , satisfying $\text{IoU} \geq 0.7$ requires $\delta \leq 0.18\ell$, where ℓ is the ground-truth span length. With one-second audio features, spans shorter than about 5.7 s therefore cannot tolerate even a one-frame shift. Tight-IoU localization is thus particularly sensitive to proposal coverage and boundary placement for short moments.

Effective AMR must therefore preserve precise local timing while retaining sufficient long-range context.

Localization quality also depends on how audio and text are aligned along the temporal axis. A query may refer to a brief acoustic event, a transition near a moment boundary, or the broader content of an extended scene, making alignment at a single temporal granularity insufficient. In addition, DETR-based moment retrieval models generate candidates using a fixed set of decoder queries [10]–[12], which may under-cover relevant regions when several short moments match the same query. We address these limitations through two complementary principles: temporal-scale- and context-aware localization, and audio–text alignment at the frame, boundary, and span levels. The resulting frame-level saliency signal is also converted into an auxiliary proposal stream at inference, extending coverage beyond the fixed decoder slots.

Following these principles, we develop a compact QD-DETR-based system using a single frozen M2D-CLAP [13] encoder. The resulting four cumulative systems were submitted to DCASE 2026 Task 6 and described in our technical report [14], with our best single model ranking joint sixth among 21 teams. We further evaluate the design through cumulative ablations, multi-seed experiments, and analyses by span duration and moment count. Our main contributions are as follows:

- We present a compact single-encoder AMR framework that jointly improves temporal localization and multi-granularity audio–text alignment.
- We introduce zero-training saliency reuse, which converts a learned frame-level alignment signal into an additional temporal proposal source at inference without additional parameters, objectives, or optimization for proposal generation.
- We provide statistically controlled and factorized analyses of our cumulative design, and show that the largest inference-stage gains occur for short, many-moment queries where proposal coverage and boundary precision are tightly coupled.

2. RELATED WORK

2.1. DETR-based moment retrieval.

Moment-DETR [15] formulated temporal grounding as direct set prediction, replacing hand-crafted proposal generation with bipartite matching. QD-DETR [10] strengthened query-conditioned cross-modal interaction, while subsequent models such as UVCOM [16], CG-DETR [17], and TR-DETR [12] further improved multimodal fusion and retrieval–highlighting interaction. These models demonstrate the effectiveness of decoder-query-based localization, but their fixed proposal budget can limit coverage when many short relevant moments occur in the same recording. Our framework retains the compact QD-DETR formulation while explicitly addressing both boundary precision and proposal coverage.

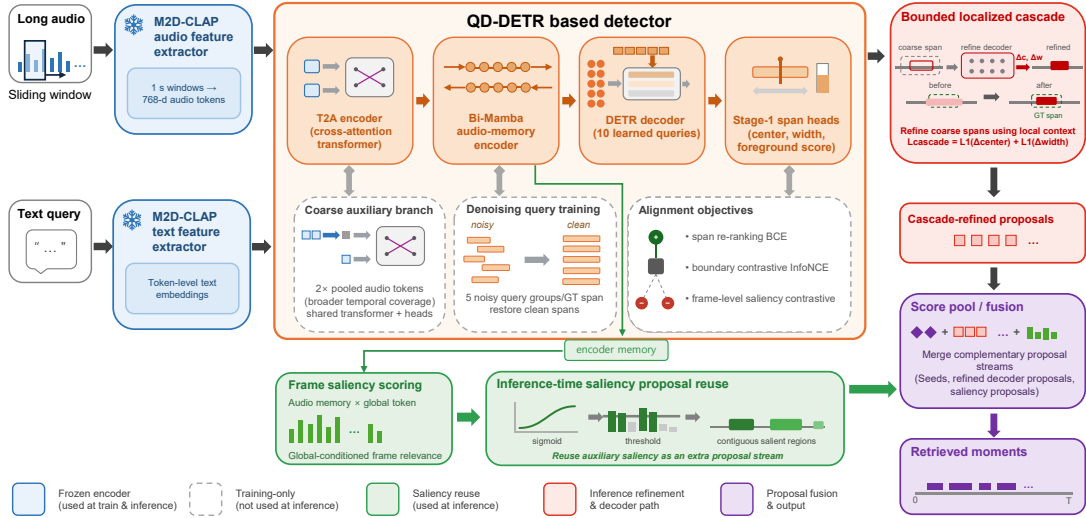


Fig. 1: Overview of the proposed AMR system. Frozen M2D-CLAP features are fed into a QD-DETR-based detector with Bi-Mamba temporal encoding. Training-only auxiliary objectives improve localization and alignment, while inference fuses cascade-refined decoder proposals with saliency-derived temporal proposals. Dashed gray modules are removed at inference.

2.2. Audio moment retrieval.

AMR was introduced by Munakata et al. [1], together with the synthetic Clotho-Moment dataset and DETR-based baselines implemented in Lighthouse [18]. CASTELLA [9] extended the task to human-annotated long-form recordings with multiple free-form queries and temporal boundaries, revealing that short moments remain particularly difficult to retrieve.

2.3. Temporal modeling and audio-text alignment.

Temporal localization has been improved through multi-scale supervision, denoising-based detection training [19], long-sequence modeling with state-space models [20], and iterative refinement [21], [22]. Moment-retrieval methods also exploit dense alignment signals such as frame-level saliency [10], retrieval-highlight reciprocity [12], and saliency-guided attention [23]. Dense relevance curves have further been converted into temporal proposals in action localization [24], [25], while concurrent AMR systems explore related boundary-aware, Mamba, saliency, and refinement mechanisms [26]–[32]. Our framework combines temporal-scale localization with frame-, boundary-, and span-level audio-text alignment, and reuses the trained saliency signal as a parallel proposal stream without additional proposal-generation training.

3. PROPOSED METHOD

3.1. Overview and base detector

Figure 1 summarizes the QD-DETR-based detector, its training-only localization and alignment modules, the localized cascade, and the saliency-derived inference proposal stream.

Following the feature-extraction pipeline of the original AMR framework [1], we divide the input audio $\mathbf{x}$ into one-second sliding windows and represent it as a sequence of window-level features. We replace its original feature extractor with frozen M2D-CLAP audio and text encoders, obtaining audio features $\mathbf{A} \in \mathbb{R}^{T \times d_f}$ and token-level query features $\mathbf{Q} \in \mathbb{R}^{L \times d_f}$ for the text query $\mathbf{q}$, where $d_f = 768$.

Following QD-DETR, a text-to-audio cross-attention encoder first conditions the audio tokens on the query tokens. We replace the

original audio self-attention encoder with a single-layer bidirectional Mamba encoder, which processes the query-conditioned sequence to aggregate temporal context over the entire recording. A conditional DETR decoder then maps $N_q = 10$ learned queries to candidate moments. For the i -th decoder query, the prediction heads output a normalized temporal span

$$\hat{\mathbf{y}}_i = (\hat{c}_i, \hat{w}_i), \quad (1)$$

represented by its center $\hat{c}_i$ and width $\hat{w}_i$, together with a foreground probability $\hat{p}_i$.

The predicted moments are assigned to the ground-truth moments through Hungarian bipartite matching. The base detection objective combines foreground classification, ℓ_1 span regression, and generalized intersection-over-union losses:

$$\mathcal{L}_{\text{det}} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\ell_1} \mathcal{L}_{\ell_1} + \lambda_{\text{gIoU}} \mathcal{L}_{\text{gIoU}}. \quad (2)$$

3.2. Temporal-scale and context-aware localization

Coarse auxiliary supervision. We average-pool the audio sequence by a factor of two and process it with the shared detector after recomputing the temporal endpoint features. The coarse predictions are supervised using the same Hungarian detection losses as the main branch, with a reduced loss weight, and the branch is removed at inference.

Denoising query training. Inspired by DN-DETR [19], we perturb the center and width of each ground-truth span with Gaussian noise and append five noisy copies as attention-isolated decoder queries. These queries are trained to recover the corresponding clean spans and are removed at inference. To the best of our knowledge, this is the first application of denoising query training to audio moment retrieval.

Bidirectional Mamba encoding. We replace the original QD-DETR audio self-attention encoder with a single-layer bidirectional Mamba encoder. Forward and reversed scans are summed and normalized to obtain the audio memory used by the DETR decoder.

Bounded localized cascade. For each first-stage prediction, the decoder re-attends to the corresponding audio-memory region expanded

by ten frames on both sides and predicts a residual correction to its center and width:

$$\hat{\mathbf{y}}_i^{\text{ref}} = \Pi_{[0,1]}(\text{sg}(\hat{\mathbf{y}}_i) + \kappa \tanh(\Delta \mathbf{y}_i)), \quad (3)$$

where $\Pi_{[0,1]}$ denotes element-wise projection onto the normalized span range, and $\text{sg}(\cdot)$ denotes the stop-gradient operation. We set $\kappa = 0.05$ to bound the magnitude of each residual correction. The residual head is zero-initialized, and the first-stage prediction is detached during cascade training. The refined spans are supervised by the same ℓ_1 and gIoU losses as the first stage, using the first-stage Hungarian assignment.

3.3. Multi-granularity audio–text alignment

Span-level alignment. For each decoder query, we average-pool the audio memory within its predicted span and project the pooled audio and mean-pooled text features into a shared embedding space. Their cosine similarity is supervised by binary cross-entropy using the Hungarian foreground assignments. This branch is used only as a training-time alignment regularizer.

Boundary-level alignment. For each ground-truth start and end boundary, we extract the $K = 3$ frames immediately inside and outside the annotated moment. The inside frames are mean-pooled to form a boundary-positive representation, while the outside frames are retained as individual negatives. Using the projected text feature as the anchor, we optimize the boundary-contrastive loss

$$\mathcal{L}_{\text{BC}} = -\frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \log \frac{\exp(\langle \mathbf{q}, \bar{\mathbf{a}}_b \rangle / \tau)}{\exp(\langle \mathbf{q}, \bar{\mathbf{a}}_b \rangle / \tau) + \sum_{t \in \mathcal{O}_b} \exp(\langle \mathbf{q}, \tilde{\mathbf{a}}_t \rangle / \tau)}, \quad (4)$$

where $\mathcal{B}$ denotes the set of valid start and end boundaries, $\bar{\mathbf{a}}_b$ is the normalized representation pooled from the inside frames of boundary b , and $\tilde{\mathbf{a}}_t$ is the normalized representation of an outside frame $t \in \mathcal{O}_b$. The projected text representation $\mathbf{q}$ is also normalized, and τ is the temperature. This objective is used only during training.

Frame-level alignment. Following QD-DETR [10], we apply a saliency contrastive objective that aligns individual audio frames with the text query. Frames inside the annotated moments are treated as relevant, while frames outside them provide negatives. In addition to supervising dense audio–text alignment, the resulting saliency logits are retained and reused as a separate proposal source at inference, as described in Section 3.4.

3.4. Inference-time ensembling and zero-training saliency reuse

Zero-training saliency reuse. We reuse the frame-level saliency head trained by the alignment objective as an additional proposal generator without further training. Given the saliency logit z_t of frame t , we compute

$$r_t = \frac{\sigma(z_t)}{\max_u \sigma(z_u) + \epsilon}, \quad s(R) = \frac{1}{|R|} \sum_{t \in R} r_t, \quad (5)$$

where $\sigma(\cdot)$ denotes the sigmoid function. We threshold r_t at $\tau_{\text{sal}} = 0.5$ and convert each contiguous above-threshold region R into a temporal proposal scored by its mean saliency $s(R)$. The ten highest-scoring saliency proposals are retained. We term this mechanism zero-training saliency reuse: proposal generation introduces no additional parameters, objectives, or optimization beyond the already trained saliency head.

Score-level ensembling and proposal fusion. We ensemble two independently trained checkpoints selected on the development set. Each checkpoint contributes its cascade-refined decoder spans and foreground scores to a common proposal pool. Because decoder queries are not aligned across checkpoints, their span coordinates are

not averaged. The saliency-derived proposals are added to the same pool, after which all candidates are ranked by their associated scores and the top ten are returned as the final retrieved moments.

4. EXPERIMENTAL SETUP

We follow the DCASE 2026 Task 6 data splits and evaluation protocol [33]. Each model is pretrained on Clotho-Moment [1] for 200 epochs and fine-tuned on CASTELLA [9] for another 200 epochs. We use AdamW with learning rates of 1×10^{-4} and 8×10^{-5} for the two stages, respectively, a batch size of 32, and a single NVIDIA RTX 3090 GPU. The frozen M2D-CLAP encoder extracts 768-dimensional features from one-second audio windows.

We use $\lambda_{\ell_1} = 10$, $\lambda_{\text{gIoU}} = 1$, and $\lambda_{\text{cls}} = 4$, with a background weight of 0.1, saliency weight of 1, coarse-branch weight of 0.25, span re-ranking weight of 0.5, boundary-contrastive weight of 0.2, and gradient clipping at 0.1.

We evaluate on the 1,347-query CASTELLA development-testing split using the official top-10 prediction protocol [33]. We report R1@0.5, R1@0.7, and mAP@[0.5:0.05:0.95], with R1@0.7 as the primary metric. The four submitted systems are additionally evaluated on the blind challenge set. For selected stages, we train multiple random seeds and report the mean and standard deviation of R1@0.7.

For adjacent system stages, we use McNemar’s exact test for R1 and paired bootstrap tests with 10^4 resamples for the remaining metrics. Multiple comparisons across the 18 tests are controlled by the Benjamini–Hochberg procedure ($q = 0.05$). In the tables, † denotes effects significant under all considered corrections, * FDR-significant effects, and (*) unadjusted-only significance.

5. RESULTS AND DISCUSSION

5.1. Overall performance and challenge evaluation

Table 1(a) summarizes the cumulative system progression. Starting from the M2D-CLAP base detector, the proposed training components progressively improve development performance, with the inference stage providing a further gain.

On the blind set, R1@0.7 improves consistently through System 3. System 3, a single model, achieved our best blind result and placed our team joint sixth among the participating teams. System 4 performs slightly below System 3 on the blind set, suggesting limited transfer of the development-selected inference settings.

5.2. Design choices and statistical significance

Table 1(b) summarizes the selected hyperparameter sweeps. To disentangle the three components introduced in System 2, we additionally perform a matched three-seed leave-one-out analysis (Table 1(c)). No individual component is decisive at this effect size. Removing Bi-Mamba lowers R1@0.7 in all three matched seeds. Including boundary alignment is beneficial in two of three seeds, although one large reversal leaves its mean effect near zero. Only the complete System 2 consistently improves over System 1 across all matched seeds. Thus, the cumulative System 2 gain is reproducible, but these experiments do not support attributing it to any single component.

The paired tests in Table 1(a) further show that each cumulative stage improves at least one evaluation aspect significantly.

5.3. Zero-training saliency reuse

At a fixed inference setting ($\alpha = 1.0$), saliency reuse improves R1@0.7 across all tested architecture–seed configurations, with consistent gains for Systems 1–3. It also outperforms alternative proposal sources that reuse the same detector at different stages or temporal resolutions,

Table 1: Summary of cumulative performance and analyses on the 1,347-query CASTELLA development-testing split and the official DCASE 2026 blind evaluation. Entries in (a) use the default seed (2023); multi-seed statistics are computed over independently trained seeds. Statistical symbols are defined in Section 4.2.

(a) Cumulative systems and blind challenge evaluation						
System	Development-testing			Multi-seed R1@0.7 $\mu \pm \sigma$ (N)	Blind evaluation	
	R1@0.5	R1@0.7	mAP		R1@0.5	R1@0.7
Official baseline (MS-CLAP)	–	13.59	–	–	–	13.56
Base detector (M2D-CLAP)	42.98	25.76	21.69	–	–	–
System 1: + coarse auxiliary supervision + span-level alignment	43.06	28.36 ^(*)	23.36*	26.13±1.74 (4)	55.93	33.90
System 2: + Bi-Mamba + boundary-level alignment + denoising	47.07*	29.55	24.30	28.47±0.80 (4)	51.41	37.85
System 3: + bounded localized cascade	48.40	31.03	25.82*	28.83±2.16 (6)	55.93	41.24
System 4: + score-level ensemble + zero-training saliency reuse	50.41	33.70*	29.16[†]	–	54.80	40.68
(b) Selected design sweeps (single seed)						
Component	Settings: R1@0.7			Sel.		
Coarse loss weight	off: 25.76; 0.25: 27.10 ; 0.50: 23.61			0.25		
Denoising noise σ	0.03: 26.50; 0.05: 29.55 ; 0.10: 27.62			0.05		
Boundary loss weight	0.1: 26.06; 0.2: 28.36 ; 0.3: 27.91			0.2		
Cascade bound κ	0.03: 29.40; 0.05: 31.03 ; 0.10: 28.73			0.05		
Saliency threshold τ_{sal}	0.3: 32.07; 0.5: 32.74 ; 0.7: 29.40			0.5		
(c) System 2 factorization (three matched seeds)						
Configuration	R1@0.7		Δ vs. S2	S2 wins		
System 1	26.75±1.50		−1.78±0.54	3/3		
– Bi-Mamba	26.51±1.09		−2.03±1.63	3/3		
– boundary alignment	28.63±2.49		+0.10±2.87	2/3		
System 2	28.53±0.97		–	–		
(d) Analysis by span duration and moment count						
R1@0.7 by ground-truth span duration						
Duration	Base	S1	S2	S3	S4	
Short < 3 s (365)	6.03	9.86	11.78	12.88	21.92	
Medium 3–10 s (534)	23.97	24.53	29.03	32.58	32.40	
Long ≥ 10 s (448)	43.97	47.99	44.64	43.97	44.87	
All (1,347)	25.76	28.36	29.55	31.03	33.70	
System 4 – base gain by moment count						
Moment count	Gain					
1 (573)	+4.19					
2–3 (404)	+6.93					
≥ 4 (370)	+14.87					
(e) Zero-training saliency-reuse analysis						
Setting	Result / test					
Proposal-source control: R1@0.7						
Decoder proposals only	31.03					
+ Saliency proposals	32.37					
+ Intermediate decoder proposals	22.42					
+ Coarse-branch proposals	30.81					
Fusion and calibration controls						
Saliency reuse only	+1.34 (McNemar $p = .027$)					
Saliency reuse, Δ mAP	+2.41 [†] ($p < 10^{-12}$)					
Per-stream isotonic calibration	+2.67 R1@0.7 ($p = .002$)					
Rank interleave	−1.34 R1@0.7					
All saliency-reuse settings are fixed across configurations.						

Table 2: R1@0.7 outcome transitions from System 3 to System 4, reported as numbers of queries. The last column analyzes only newly correct queries.

Duration	Incorrect	Correct	Saliency reuse -proposed
	→ Correct	→ Incorrect	
Short < 3 s	49	16	38 (78%)
Medium 3–10 s	39	40	5 (13%)
Long ≥ 10 s	24	20	0 (0%)

indicating that the dense saliency signal provides complementary candidates rather than merely increasing proposal count.

Because saliency and decoder scores are not calibrated to the same scale, we further analyze their reliability. At matched high-score ranges, saliency proposals are more likely to satisfy $\text{IoU} \geq 0.7$, despite decoder scores being more concentrated near one. Per-stream isotonic calibration provides a further significant improvement, while rank interleaving degrades R1@0.7. These controls indicate that the saliency scores contain useful ranking information and that the reported benefit is not an artifact of adding extra proposals.

5.4. Performance by span duration and moment count

Table 1(d) shows that retrieval remains strongly duration-dependent. The trained localization stages primarily improve medium-length moments, whereas the inference stage produces its largest gain on short moments. The improvement also grows with the number of ground-truth moments, consistent with fixed decoder queries becoming increasingly coverage-limited.

To distinguish proposal coverage from boundary refinement, we further analyze queries whose top-1 correctness changes from System 3

to System 4 (Table 2). For short moments, most newly correct predictions are supplied by the saliency stream, and many cases lack a sufficiently precise candidate among the original decoder proposals. Because the IoU tolerance of these spans is below one feature frame, proposal coverage and boundary precision become tightly coupled at the one-second resolution: a candidate displaced by a single frame may already fail the target IoU threshold. For medium moments, the inference stage mainly reorders existing candidates, while the trained localization stages account for most of the improvement.

Overall, the analysis supports the interpretation that saliency reuse primarily benefits short, many-moment queries by supplying complementary frame-level candidates rather than by continuously refining decoder boundaries.

6. CONCLUSION

We presented a compact QD-DETR-based AMR framework that combines temporal-scale modeling with frame-, boundary-, and span-level audio–text alignment. Zero-training saliency reuse converts an auxiliary alignment head into a complementary proposal stream without additional proposal-generation training or parameters. Analysis shows that the inference stage primarily benefits short, many-moment queries by supplying complementary frame-level candidates.

7. ACKNOWLEDGMENTS

This work was supported by the Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (RS-2025-02263945, HRD Program for Industrial Innovation).

REFERENCES

- [1] H. Munakata, T. Nishimura, S. Nakada, and T. Komatsu, “Language-based audio moment retrieval,” in *Proc. IEEE ICASSP*, 2025.
- [2] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *Proc. IEEE ICASSP*, 2017.
- [3] C. D. Kim, B. Kim, H. Lee, and G. Kim, “AudioCaps: Generating captions for audios in the wild,” in *Proc. NAACL-HLT*, 2019.
- [4] A. Mesaros, T. Heittola, T. Virtanen, and M. D. Plumbley, “Sound event detection: A tutorial,” *IEEE Signal Processing Magazine*, vol. 38, no. 5, pp. 67–83, 2021.
- [5] S. Adavanne, A. Politis, J. Nikunen, and T. Virtanen, “Sound event localization and detection of overlapping sources using convolutional recurrent neural networks,” *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 1, pp. 34–48, 2019.
- [6] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, “PANNs: Large-scale pretrained audio neural networks for audio pattern recognition,” *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 28, pp. 2880–2894, 2020.
- [7] A. S. Koepke, A.-M. Oncescu, J. F. Henriques, Z. Akata, and S. Albanie, “Audio retrieval with natural language queries: A benchmark study,” *IEEE Trans. Multimedia*, vol. 25, pp. 2675–2685, 2022.
- [8] A.-M. Oncescu, J. F. Henriques, and A. S. Koepke, “Dissecting temporal understanding in text-to-audio retrieval,” in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 9535–9543.
- [9] H. Munakata *et al.*, “CASTELLA: Long audio dataset with captions and temporal boundaries,” arXiv preprint, 2026.
- [10] W. Moon, S. Hyun, S. Park, D. Park, and J.-P. Heo, “Query-dependent video representation for moment retrieval and highlight detection,” in *Proc. IEEE/CVF CVPR*, 2023.
- [11] S. Park, J. Choi, K. Baek, and H. Shim, “Momentmix augmentation with length-aware DETR for temporally robust moment retrieval,” in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2026.
- [12] H. Sun, M. Zhou, W. Chen, and W. Xie, “Tr-detr: Task-reciprocal transformer for joint moment retrieval and highlight detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 5, 2024, pp. 4998–5007.
- [13] D. Niizumi, D. Takeuchi, Y. Ohishi, N. Harada, M. Yasuda, S. Tsubaki, and K. Imoto, “M2D-CLAP: Masked modeling duo meets CLAP for learning general-purpose audio-language representation,” in *Proc. Interspeech*, 2024, pp. 57–61.
- [14] S. Choi and Y.-H. Park, “Multi-signal cascaded grounding for audio moment retrieval from long audio,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [15] J. Lei, T. L. Berg, and M. Bansal, “Detecting moments and highlights in videos via natural language queries,” in *Proc. NeurIPS*, 2021.
- [16] Y. Xiao *et al.*, “Bridging the gap: A unified video comprehension framework for moment retrieval and highlight detection,” in *Proc. IEEE/CVF CVPR*, 2024.
- [17] W. Moon, S. Hyun, S. Lee, and J.-P. Heo, “Correlation-guided query-dependency calibration for video temporal grounding,” arXiv:2311.08835, 2023.
- [18] T. Nishimura, S. Nakada, H. Munakata, and T. Komatsu, “Lighthouse: A user-friendly library for reproducible video moment retrieval and highlight detection,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Nov. 2024, pp. 53–60.
- [19] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, “DN-DETR: Accelerate DETR training by introducing query denoising,” in *Proc. IEEE/CVF CVPR*, 2022.
- [20] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. Conference on Language Modeling (COLM)*, 2024.
- [21] Z. Cai and N. Vasconcelos, “Cascade R-CNN: Delving into high quality object detection,” in *Proc. IEEE/CVF CVPR*, 2018.
- [22] S. Liu, F. Li, H. Zhang, X. Yang, X. Qi, H. Su, J. Zhu, and L. Zhang, “DAB-DETR: Dynamic anchor boxes are better queries for DETR,” in *Proc. ICLR*, 2022.
- [23] A. Gordeev, V. Dokholyan, I. Tolstykh, and M. Kuprashevich, “Saliency-guided detr for moment retrieval and highlight detection,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, March 2026, pp. 907–916.
- [24] Y. Zhao, Y. Xiong, L. Wang, Z. Wu, X. Tang, and D. Lin, “Temporal action detection with structured segment networks,” in *Proc. IEEE ICCV*, 2017.
- [25] T. Lin, X. Zhao, H. Su, C. Wang, and M. Yang, “BSN: Boundary sensitive network for temporal action proposal generation,” in *Proc. ECCV*, 2018.
- [26] Y. Kang, R. Shi, B. Yen, and K. Nakadai, “FLAM-conditioned QD-DETR with paraphrase pooling and weighted-boxes-fusion ensembling for audio moment retrieval,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [27] R. Usui, R. Fujimoto, M. Kikuchi, T. Kitao, and T. Suzuki, “Improving temporal boundary precision in audio moment retrieval,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [28] Y. Hou, A. Liu, Z. Shu, X. Guo, Y. Wu, Y. Liu, X. Xia, X. Luo, and G. Huang, “Quality- and boundary-aware cross-modal refinement for long-audio moment retrieval,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [29] Y. Xu, L. Yang, and W. Zheng, “TEF-guided and quality-aware distilled DETR for language-based audio moment retrieval,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [30] J. Kim, H. J. Jung, Y. Park, and C. Lim, “QAM-DETR system for DCASE 2026 task 6: Quality-aware mamba DETR for query-based audio moment retrieval,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [31] Ó. Calvet and D. T. Toledano, “Coarse-to-fine audio moment retrieval with temporal refinement and re-ranking,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [32] T. Ogawa, N. Ohhashi, Y. Kurisuno, M. Takenaka, and R. Miyamoto, “Enhanced audio moment retrieval approach for DCASE 2026 task 6,” DCASE 2026 Challenge, Tech. Rep., Jun. 2026.
- [33] “DCASE 2026 challenge task 6: Audio moment retrieval from long audio,” <https://dcase.community/challenge2026/task-audio-moment-retrieval-from-long-audio>, 2026.